

TSFN-Based Early Fake News Detection Framework using Social Propagation Analysis

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Problem Statement

Fake news spreads rapidly on social media platforms and misleads users.

Challenges:

- ▶ Content alone is insufficient to detect fake news
- ▶ Fake news spreads differently than real news
- ▶ Social influence plays a major role

Goal: Build a hybrid system combining content + social propagation.

Objective

Primary Goal: Develop an advanced fake news detection system by integrating semantic understanding with social propagation analysis.

Key Objectives:

- ▶ Capture deep semantic meaning of news content using SBERT embeddings
- ▶ Model user credibility through Two-Stage Fusion Network (TSFN)
- ▶ Analyse how information spreads using graph-based features
- ▶ Reduce noise using First-K user sampling strategy
- ▶ Design a hybrid feature space combining content, user behaviour, and network signals
- ▶ Achieve robust and scalable classification using XGBoost

Expected Outcome: A high-accuracy, real-world adaptable fake news detection framework that outperforms traditional content-only approaches.

Dataset Used

Source: FakeNewsNet (Kaggle)

Dataset Composition: We use **four datasets** combining both political and entertainment news:

- ▶ **PolitiFact Fake** (Fake Political News)
- ▶ **PolitiFact Real** (Verified Political News)
- ▶ **GossipCop Fake** (Fake Entertainment News)
- ▶ **GossipCop Real** (Verified Entertainment News)

Total Data:

- ▶ 23,196 news articles
- ▶ Balanced mix of Fake and Real samples

Usage in our project:

- ▶ Train SBERT embeddings
- ▶ Train TSFN (trust scoring)
- ▶ Train final XGBoost classifier

Base Paper: TSFN-based Fake News Detection

Paper: Enhancing Fake News Detection using User Behaviour

Core Idea:

- ▶ Model **user behaviour** instead of only text
- ▶ Introduce **TSFN score (0–1)** for each user

TSFN Formula:

$$TSFN = \frac{n_{sf}}{n_{sf} + n_{sr} + n_f}$$

- ▶ n_{sf} : fake n_{sr} : real n_f : fact-check

Features Used:

- ▶ TF-IDF (word frequency)
- ▶ Word embeddings (semantic meaning)
- ▶ Emotional signals (user behaviour)

Models:

- ▶ Random Forest Regressor (TSFN)
- ▶ Random Forest Classifier (Fake/Real)

Base Paper: Pipeline & Limitations

Pipeline:

- ▶ User Tweets
- ▶ Feature Extraction
- ▶ TSFN Estimation
- ▶ Statistical Aggregation
- ▶ Final Classification

Limitations:

- ▶ Requires **large number of tweets**
- ▶ No **graph/network** features
- ▶ No **early detection (Top-K users)**
- ▶ Uses **traditional embeddings (TF-IDF)**

Observation:

- ▶ Works well only when **sufficient user data is available**
- ▶ Not suitable for **real-time fake news detection**

Our Proposed Improvements (Part 1)

Motivation:

- ▶ Improve **early fake news detection**
- ▶ Reduce dependency on **large tweet history**
- ▶ Incorporate **social + network behaviour**

1. Advanced Text Representation

- ▶ Replaced TF-IDF with **SBERT embeddings (384-d)**
- ▶ Captures deeper **semantic meaning**

2. Hybrid TSFN Model

- ▶ TSFN $\rightarrow$ XGBoost (learned score)
- ▶ TSFN $\rightarrow$ Cosine similarity (prototype-based)

Final TSFN:

$$TSFN = 0.7 \cdot TSFN_1 + 0.3 \cdot TSFN_2$$

3. Graph Features Added

- ▶ User degree, connectivity
- ▶ Captures **propagation patterns**

Our Proposed Improvements (Part 2)

4. Social Features Added

- ▶ Followers, Likes, Retweets
- ▶ Measures **user influence**

5. Early Detection using Top-K Users

- ▶ Select first K users (early spreaders)
- ▶ Enables **early-stage detection**
- ▶ Reduces noise and computation

6. Improved Classification Model

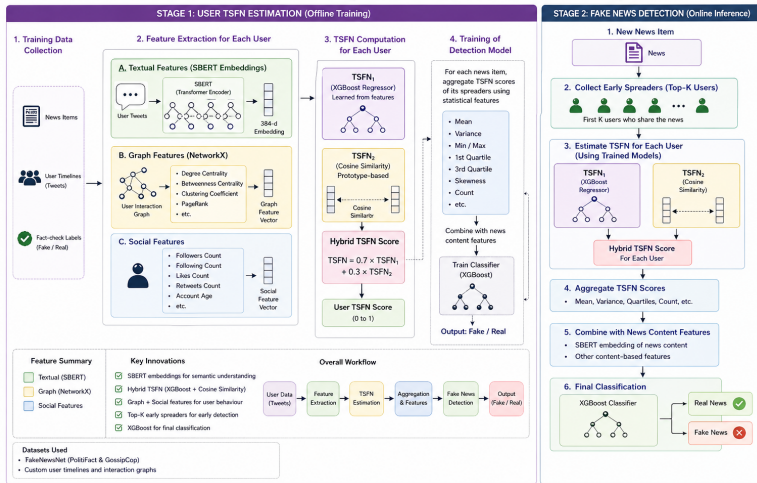
- ▶ Uses **XGBoost** instead of Random Forest
- ▶ Better performance on structured features

Overall Improvements:

- ▶ Combines **text + user behaviour + network**
- ▶ Works with **less data**
- ▶ More **robust and scalable**

Overall Architecture of Proposed System

Architecture of Our Proposed Fake News Detection System



Architecture Explanation (1/2)

1. Text Representation (SBERT)

- ▶ Converts news/tweets into **384-dimensional semantic vectors**
- ▶ Captures meaning beyond keywords (better than TF-IDF)

2. User & Social Modelling

- ▶ Generate users (or use real users if available)
- ▶ Extract:
 - ▶ Followers, likes, retweets
 - ▶ Graph structure (user connections)

3. TSFN Computation (Core Innovation)

- ▶ $TSFN_1$: XGBoost $\rightarrow$ learns fake/real patterns
- ▶ $TSFN_2$: Cosine similarity $\rightarrow$ compares behaviour
- ▶ Final:

$$TSFN = 0.7 \times TSFN_1 + 0.3 \times TSFN_2$$

Architecture Explanation (2/2)

4. Early Detection using Top-K Users

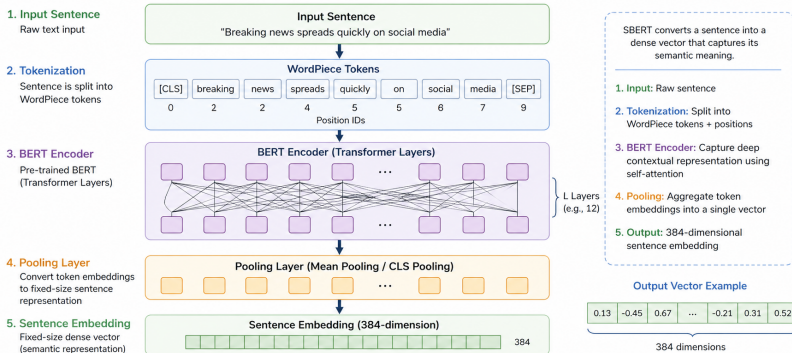
- ▶ Only first K users are selected
- ▶ Focuses on early spreaders of news
- ▶ Reduces noise and improves speed

5. Feature Extraction & Classification

- ▶ Extract statistical features:
 - ▶ Mean, variance, standard deviation
 - ▶ Min, max, range
 - ▶ Burstiness = $\text{variance} / \text{mean}$
- ▶ Combine with social + graph features
- ▶ Final prediction using **XGBoost classifier**

SBERT Architecture

SBERT Architecture (Sentence-BERT)



Key Idea:

- ▶ Converts sentence into 384-dimensional embedding
- ▶ Uses BERT encoder + pooling layer
- ▶ Captures semantic meaning of text

TSFN Computation (Mathematical View)

Goal: Estimate trustworthiness of content/user

1. $TSFN_1$ — XGBoost

- ▶ Input: SBERT embedding $\mathbf{x} \in \mathbb{R}^{384}$
- ▶ Learns mapping: $TSFN_1 = f(\mathbf{x})$
- ▶ Output: score in $[0, 1]$

2. $TSFN_2$ — Cosine Similarity

- ▶ Compare with dataset centre $\mathbf{c}$

$$TSFN_2 = \frac{\mathbf{x} \cdot \mathbf{c}}{||\mathbf{x}|| ||\mathbf{c}||}$$

3. Final TSFN (Fusion)

$$TSFN = 0.7 \cdot TSFN_1 + 0.3 \cdot TSFN_2$$

Interpretation:

- ▶ High $\rightarrow$ Real/trustworthy Low $\rightarrow$ Fake tendency

XGBoost Working (Step-by-Step)

Input:

- ▶ SBERT embedding (384 features), Label (Fake = 1, Real = 0)

Step 1: Initialize

- ▶ Start with initial prediction (mean value)

Step 2: Decision Trees

- ▶ Split features: $x_5 > 0.23$, $x_{120} < -0.5$
- ▶ Learns patterns from embeddings

Step 3: Boosting

- ▶ Each tree corrects previous errors, Improves prediction iteratively

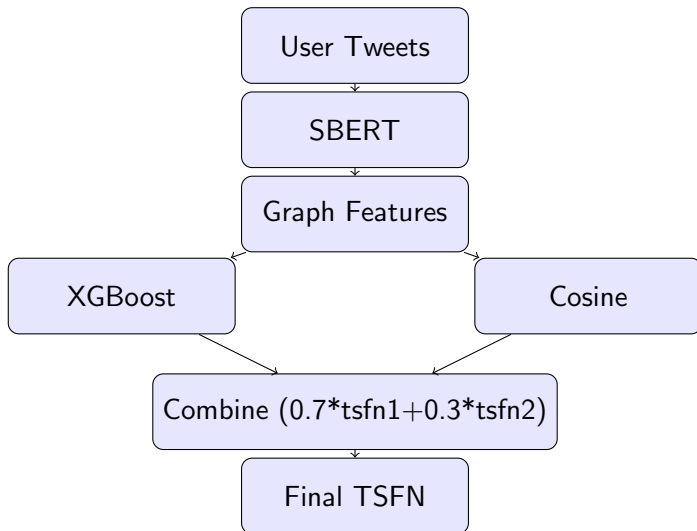
Step 4: Final Prediction

$$TSFN_1 = \sum_{k=1}^K f_k(\mathbf{x})$$

Output:

- ▶ Combined tree output $\rightarrow$ final trust score ($TSFN_1$)

Architecture



Top-K Users & TSFN Features

Top-K User Selection

- ▶ Select first K users:

$$U = \{u_1, u_2, \dots, u_K\}, \quad K = 5$$

- ▶ Why?
 - ▶ Early users $\rightarrow$ informative
 - ▶ Reduces noise

TSFN Scores

$$T = \{t_1, t_2, \dots, t_K\}$$

Statistical Features

$$\mu = \frac{1}{K} \sum t_i$$

$$\sigma^2 = \frac{1}{K} \sum (t_i - \mu)^2$$

$$B = \frac{\sigma^2}{\mu + \epsilon}$$

Why?

- ▶ Mean $\rightarrow$ trust level
- ▶ Variance $\rightarrow$ disagreement
- ▶ Burstiness $\rightarrow$ viral spikes

Social & Graph Features

Social Features

- ▶ Avg Followers
- ▶ Avg Likes
- ▶ Avg Retweets

Why Social?

- ▶ Measures influence
- ▶ Captures engagement
- ▶ Detects viral behaviour

Graph Feature

$$\text{AvgDeg} = \frac{1}{K} \sum d(u_i)$$

Why Graph?

- ▶ Models propagation structure
- ▶ Identifies key users
- ▶ Captures spread strength

Confusion Matrix & Performance

```
from sklearn.metrics import confusion_matrix, classification_report  
  
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))  
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
... Confusion Matrix:  
[[3354 120]  
 [ 743 423]]  
  
Classification Report:  
              precision    recall  f1-score   support  
  
    0           0.82       0.97       0.89       3474  
    1           0.78       0.36       0.50       1166  
  
 accuracy          0.80          0.66          0.81       4640  
 macro avg          0.80          0.66          0.69       4640  
 weighted avg          0.81          0.81          0.79       4640
```

Key Insights:

- ▶ High accuracy for Real news detection
- ▶ Some Fake news misclassified
- ▶ Fake news patterns are harder to detect

Limitations

- ▶ Synthetic users used (no real social data)
- ▶ Limited social features (followers, likes, retweets)
- ▶ Graph is approximated, not real network
- ▶ Lower performance on fake news detection

Future Work

- ▶ Use real-time social media data (APIs)
- ▶ Improve graph modelling (real networks, GNNs)
- ▶ Better early detection using dynamic Top-K
- ▶ Extend to multi-modal data (images, videos)

Conclusion

- ▶ Hybrid fake news detection using text + social + graph
- ▶ SBERT + TSFN improves detection performance
- ▶ Top-K users enable early-stage detection
- ▶ Achieved **86.7% accuracy**
- ▶ BasePaper **81.4% accuracy**

Key Insight: Combining content + user behaviour + propagation improves results

Thank You

Thank You!

Questions?

**TSFN-Based Early Fake News Detection Framework using Social
Propagation Analysis**